

Feature 2 (generate a quiz based on weak topics):

Problem: After students take the quiz, based on the questions they got wrong, if they have to retake the whole quiz, they are less likely to improve because they are doing problems they are already strong with, and also will have little exposure to their specific weak areas since the quiz is based on the full study guide.

Target Users: STEM students who want to practice or test their knowledge in the areas where they are weak.

Proposed Solution: If there is an option to let users generate a quiz based solely on their weak areas, they are more likely to improve those areas because they spend more time working on them and can also narrow down additional weak areas.

Goals:

- **Primary metric:**
 - The percentage score on the main quiz increases after they take the quiz based solely on their weak areas.
- **Guardrails:**
 - The score doesn't drop below what it was when they first took the quiz.

Scope of the solution: Let students generate a personalized quiz based on the weak areas from the quiz they took.

Out of scope: A generic quiz that is not based on the things they did wrong.

MVP / Functional Requirements

Core flow:

- [P0] After quiz submission, the system tags each incorrectly answered question with its topic and marks topics with wrong answers as "weak topics."
- [P0] If weak topics exist, a "Generate quiz from weak areas" option appears below the quiz results.
- [P0] On click, the system retrieves indexed material chunks limited to the tagged weak topics.
- [P0] The system generates and displays a quiz scoped to those weak topics only.

Edge case handling:

- [P0] If the student scores perfectly, the generated quiz option does not appear; a positive confirmation state is shown instead.
- [P1] If exactly one weak topic exists, the system generates a condensed 3-question quiz.
- [P1] If the entire quiz is answered incorrectly, the student gets the option to retake the complete quiz, which will be based on the main study guide but will have different questions.
- [P1] If quiz generation fails on retry, the student is redirected to the weak-area study guide (existing feature) with a message: "Quiz generation is temporarily unavailable, here's a study guide on these topics instead." Failure is logged for admin review.

Feedback/telemetry:

- [P0] Each generated quiz includes an "Is the grading correct?" flag, visible to the admin.
- [P0] Admin can monitor flag rate per generation batch; grading logic is flagged for review if flag rate exceeds 10%.

Assumptions:

- The quizzes based on the same materials are of the same difficulty.